

Dylan Bailes

MECHANICAL ENGINEERING | CONTROLS | EMBEDDED SYSTEMS

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U.S. citizen | Eligible for security clearance

UC San Diego mechanical engineering M.S. candidate (June 2027) with AeroVironment SUAS experience and hands-on work in electromechanical design, controls, embedded systems, and robotics. Connects modeling and simulation with firmware, fabrication, integration, and testing.

EDUCATION

M.S. Mechanical Engineering | University of California, San Diego Expected Jun 2027

Graduate focus: Optimal Linear Control, Nonlinear Control, Parametric Identification of Systems, Embedded Systems

B.S. Mechanical Engineering | University of California, San Diego Jun 2026

Controls and Robotics specialization | GPA 3.62/4.00 | Provost Honors | Degree requirements completed in three years

Selected coursework: Linear Control Design, Intro to Autonomous Vehicles, Dynamics and Control of Aerospace Vehicles, Solid Mechanics I and II, Computational Methods for Engineering and Design, Linear Circuits

TECHNICAL SKILLS

Software and controls: Python, C++, MATLAB, ROS2, Linux; Kalman filtering, state-space control

Embedded and electronics: STM32, HAL, direct register access, I2C, UART, ESP32, Raspberry Pi, KiCad PCB design

Design and fabrication: SolidWorks, Fusion 360, ANSYS FEA and Maxwell, GD&T, CNC, manual mill and lathe

EXPERIENCE

Mechanical Engineering Intern | AeroVironment | Simi Valley, CA Jun-Aug 2025

- Designed and prototyped a motor-driven SUAS mechanism under tight mass, packaging, and load constraints; completed prototype and initial design for one subassembly.
- Developed physics-based motor and spring models; a 99.5% confidence GD&T/tolerance analysis showed insufficient spring-force margin at the low end of the tolerance range.
- Created SolidWorks/FEA automation that refined meshes near holes and tight corners, fit stress-convergence curves, and estimated asymptotic peak stress.

Mechanical Lead | Robodex FIRST Robotics Competition | Granada Hills, CA Aug 2019-Jun 2023

- Led mechanical design of two award-winning FRC robots, the team's first awards in more than 20 years; fabricated 300+ parts across CAD, CNC, and manual machining.
- Logged 1,000+ hands-on hours and integrated mechanisms with electronics and programming teams; taught an elementary-school robotics curriculum.

Audio Engineer / Intern | RK Media | Thousand Oaks, CA Jun 2022-Oct 2024

- Wrote a file-sorting tool to automate the team's media organization workflow.

PROJECTS

Multi-Chamber Camera Bioreactor UC San Diego | 2026

- Designed, built, programmed, simulated, and tested the bioreactor's magnetic-field subsystem; measured 4.98 mT at coil center and less than 1% variation across the well footprint.
- Simulated electromagnet geometry in ANSYS Maxwell; designed the drive/control electronics, two KiCad PCBs, and STM32 firmware with I2C and UART interfaces.
- Debugged option-byte settings and used direct register access when HAL failed; tested the magnetic-field subsystem on a working prototype setup.

Autonomous Car Racing UC San Diego | Spring 2026

- Built and integrated a ROS2 autonomous race-car platform with camera, LiDAR, and GPS sensing alongside three ECE teammates.
- Coordinated interfaces among distinct deep-learning perception, Kalman-filter state estimation, and state-space control functions.

Flexible-Shaft Torque Control Testbed In progress | 2026

- Completed mechanical design and controller code for a compliant music-wire torque testbed comparing PID, MPC, and learned MPC; awaiting parts for hardware validation.